

# Function Reference

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## Function reference

Complete reference for the four worksheet functions exposed by the US Census Calc add-in. For setup (API key, build, install) see *INSTALL.md*; for a worked example spreadsheet see `demo/Census-Demo.ods` (generated by `tools/build_demo.py`).

All formulas below use Calc's **semicolon** argument separator. Arguments in `[brackets]` are optional and may be omitted entirely (trailing optional arguments can simply be left off the end of the formula).

### Shared behavior

- **year** accepts either a number or a text cell (e.g. `2022` or `"2022"`).
- **dataset** is the API's dataset path, e.g. `"acs/acs5"` (ACS 5-year estimates), `"acs/acs1"` (ACS 1-year), or `"dec/pl"` (decennial redistricting data). Leading/trailing slashes are stripped automatically.
- **geography** maps to the API's `for=` clause, e.g. `"state:*` (every state), `"state:06"` (California only), or `"county:*` (every county — combine with `in_geography` to scope to one state).
- **in\_geography** (where present) maps to the API's `in=` clause, e.g. `"state:06"`. Required whenever `geography` is below the state level (county, tract, block group, ...).
- **api\_key** (where present) overrides the `CENSUS_API_KEY` environment variable for that one call. Type it literally or reference a cell, e.g. `$B$1`. If both are omitted, the request is sent **without a key** — the Census API allows this for low-volume, non-commercial use (see *INSTALL.md*).
- **Caching**: every distinct request URL is cached for the life of the LibreOffice session, so recalculating a sheet does not re-hit the API. Restart LibreOffice to clear the cache.
- **Errors**: invalid input, an API error response, or a network failure all surface as a genuine Calc error value in the cell (e.g. `Err:502`), never as a raw exception string or a silent wrong answer.
- **Array formulas**: `CENSUS_GET` and `CENSUS_DATASETS` can return more than one row. LibreOffice has no dynamic spill, so select an output range large enough for the result, type the formula, and confirm with **Ctrl+Shift+Enter** (or tick **Array** in the Function Wizard). A single-cell entry shows only the top-left value of the result.

### CENSUS\_GET

Runs a Census API data query and returns the full result table.

```
CENSUS_GET(year; dataset; variables; geography; [in_geography]; [api_key])
```

| Argument             | Type           | Required | Description                                      |
|----------------------|----------------|----------|--------------------------------------------------|
| <code>year</code>    | number or text | yes      | The data year, e.g. <code>2022</code> .          |
| <code>dataset</code> | text           | yes      | The dataset path, e.g. <code>"acs/acs5"</code> . |

| Argument                  | Type | Required | Description                                                                                                                              |
|---------------------------|------|----------|------------------------------------------------------------------------------------------------------------------------------------------|
| <code>variables</code>    | text | yes      | Comma-separated variable codes, e.g. <code>"NAME,B01003_001E"</code> . Mapped to the API's <code>get=</code> parameter.                  |
| <code>geography</code>    | text | yes      | The <code>for=</code> clause, e.g. <code>"state:*)"</code> . May list several specific geographies, e.g. <code>"state:06,36,48"</code> . |
| <code>in_geography</code> | text | no       | The <code>in=</code> clause, e.g. <code>"state:06"</code> .                                                                              |
| <code>api_key</code>      | text | no       | Overrides <code>CENSUS_API_KEY</code> for this call.                                                                                     |

**Returns:** a rectangular array. Row 1 is the column headers exactly as the API returns them (variable codes, then geography identifier columns such as `state` or `county` ). Each subsequent row is one geography's values. Every cell is returned as **text**, matching the API's own JSON-array-of-arrays shape with minimal parsing — wrap numeric columns in `VALUE()` if you need them as numbers for arithmetic.

**Errors:** `dataset` / `variables` / `geography` empty, an unknown dataset or variable, a malformed geography clause, or a network failure all raise a Calc error value. A query that legitimately returns zero rows (nothing matched) is also an error, since a table with no data is never useful in a sheet.

#### Examples:

```
=CENSUS_GET(2022; "acs/acs5"; "NAME,B01003_001E"; "state:*)"
  -> 53 rows (header + 52 states/territories): name, total population

=CENSUS_GET(2022; "acs/acs5"; "NAME,B01003_001E"; "state:06,36,48")
  -> 4 rows: header + California, New York, Texas

=CENSUS_GET(2022; "acs/acs5"; "NAME,B01003_001E"; "county:*"; "state:06")
  -> 59 rows: header + every California county (in_geography scopes to CA)

=CENSUS_GET(2020; "dec/pl"; "NAME,P1_001N"; "state:*)"
  -> 2020 decennial redistricting total population by state
```

## CENSUS\_VALUE

Returns a single scalar value for one variable at one specific geography.

```
CENSUS_VALUE(year; dataset; variable; geography; [in_geography]; [api_key])
```

| Argument                  | Type           | Required | Description                                                                                                                         |
|---------------------------|----------------|----------|-------------------------------------------------------------------------------------------------------------------------------------|
| <code>year</code>         | number or text | yes      | The data year, e.g. <code>2022</code> .                                                                                             |
| <code>dataset</code>      | text           | yes      | The dataset path, e.g. <code>"acs/acs5"</code> .                                                                                    |
| <code>variable</code>     | text           | yes      | A single variable code, e.g. <code>"B01003_001E"</code> .                                                                           |
| <code>geography</code>    | text           | yes      | The <code>for=</code> clause identifying <b>one specific geography</b> , e.g. <code>"state:06"</code> — not a wildcard (see Notes). |
| <code>in_geography</code> | text           | no       | The <code>in=</code> clause, e.g. <code>"state:06"</code> .                                                                         |

| Argument | Type | Required | Description                             |
|----------|------|----------|-----------------------------------------|
| api_key  | text | no       | Overrides CENSUS_API_KEY for this call. |

**Returns:** the requested value from the first (and normally only) data row — a **number** when it parses as one (most ACS estimates), **text** for string fields (e.g. NAME ), or an **empty cell** when the API returns a null value for that variable/geography.

**Errors:** dataset / variable / geography empty, no data row returned for the query, or the requested variable not present in the response header (e.g. a typo in the variable code) all raise a Calc error value.

**Notes:** internally this runs the same kind of query as CENSUS\_GET with variables set to just variable , then reads the **first** data row. If geography resolves to more than one row (e.g. a wildcard like "state:\*" ), you get whichever row the API happened to return first — pass a specific geography instead, or use CENSUS\_GET to pull every matching geography at once.

#### Examples:

```
=CENSUS_VALUE(2022; "acs/acs5"; "NAME"; "state:06")
  -> California

=CENSUS_VALUE(2022; "acs/acs5"; "B01003_001E"; "state:06")
  -> 39356104

=CENSUS_VALUE(2022; "acs/acs5"; "B19013_001E"; "county:075"; "state:06")
  -> median household income for San Francisco County, CA

=CENSUS_VALUE(2022; "acs/acs5"; "B01003_001E"; "state:06"; ""; "$B$1")
  -> same query, API key taken from cell B1
```

## CENSUS\_VARLABEL

Returns the human-readable label for a variable code, from the dataset's variables.json metadata.

```
CENSUS_VARLABEL(year; dataset; variable)
```

| Argument | Type           | Required | Description                                        |
|----------|----------------|----------|----------------------------------------------------|
| year     | number or text | yes      | The data year, e.g. 2022 .                         |
| dataset  | text           | yes      | The dataset path, e.g. "acs/acs5" .                |
| variable | text           | yes      | The variable code to look up, e.g. "B01003_001E" . |

**Returns:** the variable's label field (e.g. "Estimate!!Total" ), or its name if the dataset has no label for it.

**Errors:** dataset / variable empty, or the variable code not found in that dataset/year, raises a Calc error value.

**Notes:** this endpoint is metadata, not data — it never needs an API key, so there is no api\_key argument.

#### Examples:

```
=CENSUS_VARLABEL(2022; "acs/acs5"; "B01003_001E")
  -> Estimate!!Total
```

```
=CENSUS_VARLABEL(2022; "acs/acs5"; "B19013_001E")  
-> Estimate!!Median household income in the past 12 months (in 2022 inflation-adjusted dollars)
```

Pair it with `CENSUS_GET / CENSUS_VALUE` to build a self-documenting sheet: put variable codes in column A and `=CENSUS_VARLABEL(2022;"acs/acs5";A2)` in column B.

## CENSUS\_DATASETS

Lists the datasets available for a year, from the API's discovery endpoint.

```
CENSUS_DATASETS(year)
```

| Argument | Type           | Required | Description                |
|----------|----------------|----------|----------------------------|
| year     | number or text | yes      | The data year, e.g. 2022 . |

**Returns:** a two-column array. Row 1 is a `Dataset / Title` header row; each subsequent row is one available dataset's path (e.g. `"acs/acs5"` , joined from the API's `c_dataset` segments) and its title. A given year typically has ~100 datasets, so select a generous range (or a full column) when entering this as an array formula.

**Errors:** `year` empty, or no datasets listed for that year (e.g. a year far outside the API's coverage), raises a Calc error value.

**Notes:** this is the discovery endpoint, not a data query — it never needs an API key. Row order matches whatever order the API itself returns (not alphabetical); use `CENSUS_GET / CENSUS_VALUE` once you've found the dataset path you want.

### Examples:

```
=CENSUS_DATASETS(2022)  
-> spills ~100 rows: (Dataset, Title) for every 2022 dataset  
  
=CENSUS_DATASETS(2020)  
-> includes "dec/pl" ("Decennial Census: Redistricting Data (PL 94-171)")
```